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Thermodynamic assessment of the Ce–H and $CeNi_5$–H system

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ABSTRACT

Interstitial metal hydrides (MHs) have attracted considerable attention in the field of hydrogen technology, particularly in the context of storage and compression applications. Because of their minor hysteresis effects, good cyclability, activation simplicity, and high volumetric storage density, $LaNi_5$-based alloys are recognized as prominent candidates for hydrogen storage application. Additionally, the system's thermodynamic and electrochemical properties can be modified to suit the requirements of a particular application by alloying specific substituents. To ascertain the thermodynamic effects of Ce addition within $LaNi_5$, in this work the Ce–H and $CeNi_5$–H systems have been modeled with the CALPHAD method. For this reason, in this work, two different thermodynamic models have been developed and assessed using the same pressure-composition isotherms (PCIs) datasets obtained from literature and theoretical formation energies newly calculated employing periodic density functional theory (DFT). Direct comparison of the models against each other in terms of accuracy and physical plausibility revealed that extrapolation of thermodynamic properties to data-scarce regions is more reasonable with fewer model parameters and in agreement with other similar systems within the rare-earth (RE) metal-hydride class. In addition, the $CeNi_5$–H system was investigated by assessing the $(Ce)(Ni)_5(Va,H)_7$ phase model, which could accurately predict hydrogen storage properties while being compatible with previously developed $LaNi_5$–H models. Ultimately, the models developed in this study may be employed and extended to describe multi-component RE–H systems and allow for thermodynamic computations that are highly desirable for accurate predictions of hydrogen absorption/desorption properties and degradation characteristics within the $(La,Ce)Ni_5$–H metal hydride family.

1. Introduction

Metal hydrides are considered viable materials for storing hydrogen in solid-state due to their large achievable volumetric hydrogen storage capacity (100–120 kg/m^3 for intermetallic compounds) and intrinsic safety in comparison to physical storage methods such as liquefaction at cryogenic temperatures or compression at hundreds of bar [1,2]. Beyond storage, these materials find applications in areas like hydrogen gas compression and as cathode materials, e.g., in Ni-MH batteries [3,4]. Among metal hydrides, the AB_5 system (A = La, Ce, ...; B = Ni, Al, Fe, ...) has emerged as a prominent materials class of interest. This attention is attributed to the distinct advantages that AB_5 hydrides' possess over other metal hydrides (e.g, Mg, and $TiMn_2$ [4–6]), i.e., moderate operation conditions, high degree of cyclability, and good activation properties.

With the substitution of Ce in $LaNi_5$, the thermodynamic stability of the resulting hydride decreases, requiring higher H_2 pressures for its formation [7,8]. Additionally, the disproportionation mechanisms of the ternary hydride into binary RE hydrides and pure Ni are also affected by the addition of Ce [9], which makes Ce a key element for modulating RE hydrides in long-term applications. Ce is therefore a prominent element for partial A-element substitution as it raises the equilibrium pressure while maintaining most of the good sorption properties of the $LaNi_5$ compound [10].

Since hydrogen sorption properties are intrinsically related to the equilibrium between metal and hydride, thermodynamic equilibrium descriptions are fundamental to understanding and predicting operating conditions for metal-hydride systems. Recent developments in this field include the FeTi–H CALPHAD model by some of us [11] as well as thermodynamic models for equiatomic and non-equiatomic TiZrNbFeNi high-entropy alloys [12].

Available CALPHAD descriptions for the AB_5 system were made by Palumbo et al. [13] in modeling the $LaNi_5$–H system and subsequent

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work by Pinatel et al. [14], which explored the thermodynamic role of Al as a B-element substituent. Even though Ce is a prominent element for A-element substitution, to our knowledge no CALPHAD assessments of the H–Ce–Ni system have been published. Intending to fill this gap, this study provides an in-depth thermodynamic assessment of the binary Ce–H and the ternary $CeNi_5$–H systems.

2. Literature review

2.1. Binary Ce–H system

The current understanding of the Ce–H phase diagram was primarily developed using various PCI measurements from literature [15–20]. In 1997, Manchester et al. compiled the available data and drafted a first phase diagram including datapoints up to 3.2 MPa [21]. The compositional values for the hydrogen-rich and hydrogen-poor two-solid-phase equilibrium regions can be identified by determining the onset and offset of the plateaux. Under pressure constraints, the hydrogen solubility limit of the single-solid-phase region is given by its equilibrium point with the gas phase.

However, as also mentioned by the authors, these measurements often come with high uncertainty levels, which are attributed to several factors [21]:

- Inappropriate vacuum conditions — PCIs are identified by calculating relative mass changes of pressure variations, which requires a high vacuum to achieve complete desorption.
- Compositional inhomogeneities — These induce a sloping plateau that fuzzes the value of the bulk equilibrium plateau pressure and smoothen the inflections of the PCIs, which in turn hinders identifying the compositional limits of the plateaux.
- Hysteresis effects — Even though their causes are still under debate [22], they introduce large uncertainties in the determination of the bulk equilibrium pressures.

Because of these effects, it is necessary to carefully analyze the experimental setup and measurement procedure indicated in the literature. To minimize these discrepancies, the values for the plateaux pressures corresponding to the thermodynamic equilibrium of the bulk phases were chosen at the mid-point of the plateau composition and as the logarithmic average of the absorption and desorption plateaux pressure [23], while the plateaux compositional limits were determined by extrapolating the composition of the single-phase regions to the assumed equilibrium pressure.

In the following, the structural and thermodynamic characteristics of phases experimentally found in the Ce–H system are summarized.

2.1.1. dhcp, fcc, bcc, and liquid phases

The low-temperature double hexagonal-close-packed structure (dhcp) exhibits negligible solubility of hydrogen [21].

From 334 K the face-centered cubic (fcc) structure is stable with an increasing solubility that reaches 0.18 at.-% H at 1003 K [19].

The body-centered cubic (bcc) structure shows an extended peritectic mixed phase region ending at the peritectic point at 1298 K and 44 at.-% H in equilibrium with the liquid and CeH_x phase, measured by Teleffsen et al. [18].

2.1.2. CeH_x

The CeH_x phase has a fcc structure with the $Fm\bar{3}m$ space group, where the Ce atoms form a bcc-like structure while the hydrogen atoms are positioned in the octahedral and tetrahedral sites. At room temperature, the CeH_x phase shows a solubility of hydrogen between 66 and 74 at.-% H [15]. The lower solubility limit of the CeH_x phase is almost constant with temperature until the peritectic point, at which the hydrogen content reaches 63 at.-% H. Above the peritectic temperature, the solubility limit reduces at the pace of increasing temperature, reaching 58 at.-% H at the highest registered temperature of 1423 K [18].

Even though at temperatures below 273 K a tetragonal distortion of the hydride phase was reported [24,25], its analysis is outside the scope of this study and was not included in the present assessment.

Due to the challenging experimental setups when high pressure is required, only a small amount of experimental data relating to the maximum hydrogen solubility in the CeH_x hydride (hydride-gas equilibrium) was available in the literature.

While multi-phase equilibrium data can be plentifully found, internal thermochemical data is generally scarce. However, the partial molar enthalpy of hydrogen could be collected from several studies [18–20,26], in which the value of ~102 kJ/mol H is attributed to the plateau region, while showing an increase with rising hydrogen content in the CeH_x phase.

2.2. Ni–H, Ce–Ni and $CeNi_5$–H systems

For the binary Ni–H system, the latest assessment by Palumbo et al. was adopted [13] in this study.

The binary Ce–Ni system was assessed by different authors [27–30]. Fig. 1 shows the calculated Ce–Ni phase diagram employing the assessed database from Ye et al. [27]. Note that the $CeNi_5$ phase is modeled as a stoichiometric compound. The system contains seven different intermetallic phases, among which only the $CeNi_5$ intermetallic is reported to absorb hydrogen. Its hydrogen solubility is non-negligible only above 773 K, and therefore, only these datapoints were considered in this study. The $CeNi_5$ phase is stable in a hexagonal crystal structure corresponding to the $P6/mmm$ space group, which remains stable until it melts at 1608 K [31], as shown in Fig. 1.

Upon hydrogenation, the $CeNi_5$ phase undergoes a high volumetric expansion of about 30 % without the formation of a superstructure. A valence change of cerium during hydrogenation of $CeNi_5$ has been proposed to explain this phenomenon [32]. Considering interstitial hydrogen, the crystal structure of the $CeNi_5$-hydride has been described with the $P6/mmm$ space group [32]. Similar as for the $LaNi_5$-hydrides, possible hydrogen ordering cannot be excluded [33].

Experimental PCI data has been published by several authors [8,34–36]. While the pressure–temperature pairs are consistent, there are noticeable discrepancies in the plateaux composition limits. These variations can be attributed to differences in the experimental setups, with some focusing on measuring absolute hydrogen content and others on assessing weight differences. Additionally, inconsistencies in vacuum quality during different cycles can lead to incomplete dehydrogenation, as previously discussed.

Lakner et al. [36] showed a plateau split at low temperatures (252 K), which was not yet discussed or reproduced by others and is thus not included in the current assessment.

3. Computational methods

3.1. Thermodynamic modeling

In the following, the Gibbs energy expressions of the unary solid elements in their stable phases at 298.15 K were chosen as the reference state for the system (SER, Standard Element Reference), following the SGTE (Scientific Group Thermodata Europe) [37] recommendation. The Gibbs energy expression of the end-members is then given by:

$$G_m^p - \sum_i a_i {}^0G_i^{\mathrm{SER}} = \Delta H_m^f + \Delta S_m^f T, \tag{1}$$

where m is the end-member of phase p, and a_i is the relative stoichiometric coefficient of species i in the end-member compound. The terms ΔH_m^f, and ΔS_m^f represent the enthalpy and entropy of formation of the related end-member, respectively. The reference state of gaseous hydrogen (${}^0G_{H_2}^{\mathrm{SER}}$ = GH2GAST) was adopted following the work from Palumbo et al. [13].

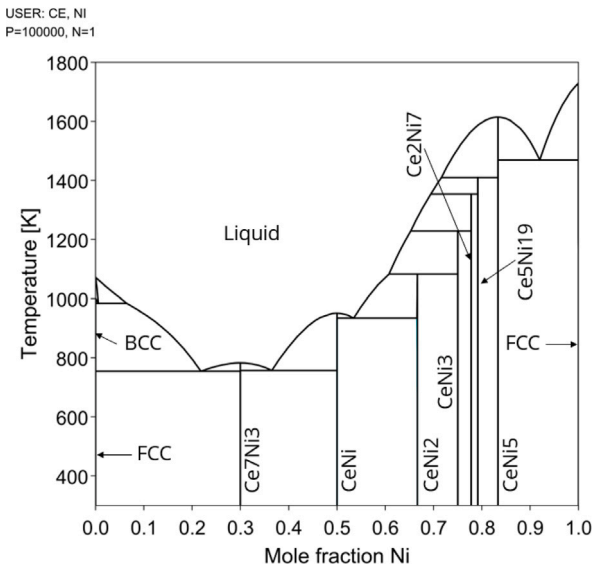


Fig. 1. Calculated Ce–Ni equilibrium phase diagram using the thermodynamic models reported by Ye et al. [27].

In general, only binary interactions in a single sublattice were considered. For the expansion of the interaction parameters, Redlich–Kister polynomials were used [38]. The Gibbs energies of the phases are given in Joules per mole of formula unit of a given phase Φ (G^{Φ}), and are expressed by:

$$G^{\Phi} = {}^{ref}G^{\Phi} + {}^{id}G^{\Phi} + {}^{ex}G^{\Phi}, \tag{2}$$

where ${}^{ref}G^{\Phi}$, ${}^{id}G^{\Phi}$ and ${}^{ex}G^{\Phi}$ are the reference, ideal mixing, and excess energy contributions to the Gibbs energy of phase Φ, respectively [39]. These contributions are modeled differently for each phase and are presented in the following sections.

3.1.1. Gas phase

The content of metallic species in the gas phase is negligible at the temperature range relevant for this study and therefore not considered in the present model. For the binary Ce–H system, the hydrogen gas was considered ideal, due to the relatively low pressures involved [40].

$$G^{gas}_{H_2} = {}^{0}G^{gas}_{H_2} + RT\ln(P/P_0), \tag{3}$$

where ${}^{0}G^{j}_{i}$ is the reference state Gibbs energy of species i in the phase j, R is the gas constant and P_0 the standard pressure of 101 325 Pa.

Due to high-pressure equilibria involved in the $CeNi_5$–H system, the non-ideal hydrogen gas model from Joubert [40] was employed for this case:

$$\begin{aligned} G^{gas}_{H_2} &= {}^{0}G^{gas}_{H_2} + RT\ln(P/P_0) - a_1 b_1 \exp(-P/b_1) \\ &- a_2 b_2 \exp(-P/b_2) - a_3 b_3 \exp(-P/b_3) \\ &- a_4 b_4 \exp(-P/b_4) - a_5 b_5 \exp(-P/b_5) + cP + d, \end{aligned} \tag{4}$$

The parameter set for a_n, b_n, c and d can be found in [40].

3.1.2. Liquid phase

The liquid phase was modeled with a single sublattice model, (Ce, H). The contributions to the Gibbs energy expression for the liquid phase (G^{liq}) are given by:

$$\begin{aligned} {}^{ref}G^{liq} &= y_H G^{liq}_H + y_{Ce} G^{liq}_{Ce} \\ {}^{id}G^{liq} &= RT\left[y_H \ln(y_H) + y_{Ce}\ln(y_{Ce})\right] \\ {}^{ex}G^{liq} &= y_H y_{Ce} \sum_{v} {}^{v}L^{liq}_{H,Ce}(y_{Ce} - y_H)^{v}. \end{aligned} \tag{5}$$

The G^{liq}_i represents the Gibbs energy expressions of the unary liquid composed of species i, and are adopted from SGTE [37]. The expression of G^{liq}_H is taken from Qui et al. [41]. y_i represents the fraction of component i in the single sublattice sites. ${}^{v}L^{liq}_{H,Ce}$ is the binary interaction term of order v between the components Ce, and H in the liquid phase [38].

3.1.3. Solution phases

The solution phases (i.e., bcc and fcc) are described with a two-sublattice model. The first and second sublattices are attributed to the metallic sites and interstitial sites, respectively. The interstitial sites may be occupied up to a hydrogen ratio of 1: x concerning metallic species, i.e. $(Ce)(H, Va)_x$, with $x = 3$ and $x = 1$ for the bcc, and fcc phases, respectively. The contributions to the Gibbs energy per formula unit of a given solution phase ϕ (G^{ϕ}) are then expressed by:

$$\begin{aligned} {}^{ref}G^{\phi} &= {}^{2}y_H\,{}^{1}y_{Ce} G^{\phi}_{Ce:H} + {}^{2}y_{Va}\,{}^{1}y_{Ce} G^{\phi}_{Ce:Va} \\ {}^{id}G^{\phi} &= xRT\left[{}^{2}y_H \ln({}^{2}y_H) + {}^{2}y_{Va}\ln({}^{2}y_{Va})\right] \\ {}^{ex}G^{\phi} &= {}^{2}y_H\,{}^{2}y_{Va}\,{}^{1}y_{Ce}\,{}^{0}L^{\phi}_{Ce:H,Va}, \end{aligned} \tag{6}$$

where Va stands for vacancies, and ${}^{m}y_i$ is the site fraction of component i in sublattice m. The ${}^{0}L^{\phi}_{Ce:H,Va}$ term is the zero-th order Redlich–Kister parameter [38] accounting for the hydrogen-vacancy interaction within the interstitial lattice. Note that for the bcc phase, the mixing was considered ideal, i.e. ${}^{0}L^{bcc}_{Ce:H,Va} = 0$.

3.1.4. CeH_x phase

The CeH_x phase exhibits similarities to other binary RE-H systems: For $x_H < 2$ hydrogen preferably fills tetrahedral sites, while for $x_H > 2$ hydrogen starts occupying octahedral sites. Hence, we described the CeH_x phase with a $(Ce)_{0.25}(H, Va)_{0.5}(H, Va)_{0.25}$ sublattice model, aligning with thermodynamic models of similar systems found in existing literature [13,42]. The contributions to the Gibbs energy expression of the CeH_x phase (G^{CeH_x}) then take the form:

$$\begin{aligned} {}^{ref}G^{CeH_x} &= {}^{3}y_H\left({}^{2}y_H G^{CeH_x}_{Ce:H:H} + {}^{2}y_{Va} G^{CeH_x}_{Ce:Va:H}\right) \\ &+ {}^{3}y_{Va}\left({}^{2}y_H G^{CeH_x}_{Ce:H:Va} + {}^{2}y_{Va} G^{CeH_x}_{Ce:Va:Va}\right) \\ {}^{id}G^{CeH_x} &= 0.5RT\left[{}^{2}y_H \ln({}^{2}y_H) + {}^{2}y_{Va}\ln({}^{2}y_{Va})\right] \\ &+ 0.25RT\left[{}^{3}y_H \ln({}^{3}y_H) + {}^{3}y_{Va}\ln({}^{3}y_{Va})\right] \\ {}^{ex}G^{CeH_x} &= {}^{2}y_H\,{}^{2}y_{Va}\left({}^{3}y_{Va}\,{}^{0}L^{CeH_x}_{Ce:H,Va:Va}\right. \\ &\left. + {}^{3}y_H\,{}^{0}L^{CeH_x}_{Ce:H,Va:H}\right) \\ &+ {}^{3}y_H\,{}^{3}y_{Va}\left({}^{2}y_{Va}\,{}^{0}L^{CeH_x}_{Ce:Va:H,Va}\right. \\ &\left. + {}^{2}y_H\,{}^{0}L^{CeH_x}_{Ce:H:H,Va}\right). \end{aligned} \tag{7}$$

The ${}^{0}L^{CeH_x}_{Ce:n:H,Va}$ and ${}^{0}L^{CeH_x}_{Ce:H,Va:n}$, where n denotes Va or H, accounts for the hydrogen-vacancy interaction in the second and third interstitial lattices, respectively, and can be temperature dependent, as discussed in Section 4.1.

3.1.5. $CeNi_5H_x$ phase

Since the $CeNi_5$ phase presents no superstructure upon hydrogenation [32], the phase can be expanded from the binary Ce–Ni system [27] by adding a sublattice to the $(Ce)(Ni)_5$ phase model to account for the hydrogen-vacancy interaction in the interstitial sites. According to recent CALPHAD assessments of the $LaNi_5$–H system a maximum hydrogen capacity of $CeNi_5H_7$ was assumed [13,43], resulting in a $(Ce)(Ni)_5(H, Va)_7$ model. The distinction between different possible space groups for the hydride is, namely $P6/mmm$, $P6$ mm and $P6_3mc$, has not been conclusively determined [32,33]. Choosing the same model as for the $LaNi_5$-hydride ensures consistency between this work and previous assessments in this system [13,43]. The contributions to

the Gibbs energy expression of the $CeNi_5H_x$ phase ($G^{CeNi_5H_x}$) are given as:

$$\begin{aligned} {}^{ref}G^{CeNi_5H_x} &= y_H G^{CeNi_5H_x}_{Ce:Ni:H} + y_{Va} G^{CeNi_5H_x}_{Ce:Ni:Va} \\ {}^{id}G^{CeNi_5H_x} &= RT\left[y_H \ln(y_H) + y_{Va}\ln(y_{Va})\right] \\ {}^{ex}G^{CeNi_5H_x} &= y_H y_{Va}\, {}^{2}L^{CeNi_5H_x}_{Ce:Ni:H,Va}(y_H - y_{Va})^2. \end{aligned} \tag{8}$$

Note that only the second-order Redlich–Kister term ${}^{2}L^{CeNi_5H_x}_{Ce:Ni:H,Va}$ was used to describe hydrogen-vacancy interaction in the sublattice of the $CeNi_5H_x$ phase.

3.1.6. Optimization procedure

The PARROT Module from Thermo-Calc Software 2022a [44] was used for the optimization of the thermodynamic model parameters. The experimental data from the PCI measurements were converted to compositional limits of the plateaux pressure and partial molar enthalpies of hydrogen.

In determining the parametrization of the models, two different approaches were chosen, named *model 1* and *model 2*. In both approaches, we made use of the same Compound Energy Formalism (CEF) [45] sublattice models for the phases, differing them only by the number and the type of model parameters coefficients. In *model 1* we aimed at the minimization of the Relative Standard Deviation (RSD) by attributing more independent coefficients for the model parameters of the CeH_x phase. Whereas for *model 2* the emphasis was the reduction of the optimizing variables of the CeH_x phase to achieve a compromise between the fitting error and reasonable phase equilibria in comparison to other RE-H models [13,42].

In summary, the models were distinguished only by how the temperature dependence of the CeH_x phase model was handled. In *model 1* the Gibbs energy expressions of the end-members of the CeH_x phase were set to be each linearly dependent with respect to temperature, while hydrogen-vacancy interaction terms were set as temperature-independent. In contrast, in *model 2* the unstable end-members of the CeH_x phase were set to be temperature-independent and their term related to the enthalpy of formation constrained to be identical. Similarly, the parameters regarding hydrogen-vacancy interaction in the third sublattice were set as temperature-independent and identical regardless of the species occupying the second sublattice.

Starting values for the parameter optimization in this study were chosen based on the values of analogous model parameters in the Sm-H and La-H assessments from Zienkevich [42] and Palumbo [13], respectively. The formation energies computed *via* DFT in this work were chosen as starting values of the parameters relating the formation enthalpies of unstable end-members of the CeH_x, and $CeNi_5H_x$ phases, (i.e., $(Ce)_{0.25}(Va)_{0.5}(Va)_{0.25}$, $(Ce)_{0.25}(Va)_{0.5}(H)_{0.25}$, and $(Ce)(Ni)_5(H)_7$ end-members), but not used as an experimental constraint during the assessments.

In general, lower weights were attributed to experimental data obtained from PCI curves in comparison to experimental data from XRD. Special attention was given to three-phase equilibria in the Ce–H system, to which the highest weights were applied during the optimization. As an initial step, invariant points that were lacking in existing literature were inserted based on the RE-H assessments from [13, 42] with the only purpose of finding appropriate initial parameters. Subsequently, these extra data points were excluded from the final optimization run.

Due to the aforementioned inconsistencies in acquiring equilibrium compositions within the $CeNi_5$ hydrogenation, only the relative values (i.e., the variation between hydrogen-rich and hydrogen-poor compositions) were implemented as data points. The absolute values of the compositional edges of the plateaux were not fixed. This led to a minimization of the number of required fitting parameters.

3.2. Ab initio calculations

DFT calculations for the unstable phases in the binary Ce–H system (i.e., $(Ce)(Va)_2(H)$ and $(Ce)(Va)_2(Va)$) and the $(Ce)(Ni)_5(H)_7$ end-member were conducted using the Quantum ESPRESSO (QE) package [46] with the local density approximation (LDA) and generalized-gradient approximation (GGA).

The Standard Solid-State Pseudopotentials (SSSP) Precision (version 1.3.0) [47] was employed, which incorporates the PBE functional with ultrasoft pseudopotentials for Ni, PAW for Ce, and norm-conserving pseudopotentials for Hydrogen. All plane waves were set to a kinetic energy cutoff of 140 Ry, following the recommendation of the SSSP for hydrogen.

Full structure relaxation was ensured, maintaining residual atomic forces below $1 \times 10^{-4}\,\mathrm{eV\,Å^{-1}}$ and an energy convergence threshold at $1 \times 10^{-5}\,\mathrm{eV}$. The Monkhorst–Pack scheme was applied with a k-point spacing of $0.04\,\mathrm{Å^{-1}}$. To compute the total energy of hydrogen, vital for evaluating formation energies, an H_2 molecule was placed in a cubic unit cell with a lattice parameter of 10 Å, ensuring sample space to prevent self-interactions due to periodic boundary conditions.

The formation enthalpies calculated at the ground-state were assumed to vary negligibly from the standard formation enthalpies [48].

4. Results and discussions

4.1. Binary Ce–H system

The DFT results are reported in Table 3. Parameter optimization results for *models 1* and *2* are reported in Table 1 and Table 2, respectively. Both approaches employ the same thermodynamic models for the phases but differ in the number and type of attributed model parameter coefficients. The key distinction between the models lies in their CeH_x phase parametrization.

In *model 1*, there were no temperature-dependent terms associated with hydrogen-vacancy interaction, i.e. it was assumed an ideal and/or regular solution of hydrogen in the interstitial lattice. Instead, temperature dependencies were modulated through the entropy of formation of the end-members (see Eq. (1) for clarity). These were considered for each of the end-members of the CeH_x phase as shown in Table 1.

In contrast, *model 2* featured a total of eight adjustable coefficients. Temperature dependence was integrated into the hydrogen-vacancy interaction parameters that were introduced via linear temperature dependence within its zero-th order Redlich–Kister term, while the entropy of formation terms were only assigned to stable end-members (Table 2 for reference).

The *model 1* assessment led to a more accurate description of the experimental points (lower RSD), as can be inferred from Fig. 2. The peritectic point was calculated with a deviation of 0.3 at.%-H and 1 K (as shown in Fig. 2(*a*)) compared to 1.6 at.%-H and 5 K obtained with the second approach (illustrated in Fig. 2(*b*)). While the fcc-bcc-CeH_x three-phase is calculated practically with the same equilibrium temperature, the two-phase equilibria compositions of the CeH_x phase show non-negligible variations between the two modeling approaches. Utilization of more adjustable coefficients resulted in better agreement with experimental PCI data regarding equilibrium composition between CeH_x, the solution phases and the liquid phase as shown in Fig. 3.

Although no critical point is experimentally observed for binary RE-H systems, it is expected that the compositional range of the plateaux pressure diminishes with rising temperatures as it is known to occur in many other binary and ternary metal-hydrogen systems [49]. This phenomenon could be better represented if more parameters were employed in the thermodynamic models. On the contrary, for the second modeling approach, the plateaux terminations result in increasing hydrogen contents over increasing temperatures.

While the use of more adjustable coefficients (*model 1*) yields better agreement concerning the measured phase's composition, as evidenced

Table 1
Assessed thermodynamic parameters for *model 1* of the Ce–H system [J/mol f.u.].

Parameter	Values	Reference
Liquid phase		
G^{liq}_{Ce}	GLIQCE	[37]
G^{liq}_{H}	+8035+25*T+2*T*ln(T) +0.5*GH2GAST	[41]
${}^{0}L^{liq}_{H,Ce}$	−49587 - 41.87*T	This work
${}^{1}L^{liq}_{H,Ce}$	46543 - 60.23*T	This work
bcc phase		
$G^{bcc}_{Ce:Va}$	GBCCCE	[37]
$G^{bcc}_{Ce:H}$	−112720 + 104.11*T + GHSERCE + 1.5*GH2GAST	This work
fcc phase		
$G^{fcc}_{Ce:Va}$	GHSERCE	[37]
$G^{fcc}_{Ce:H}$	−53676 + 392.11*T + GHSERCE + 0.5*GH2GAST	This work
${}^{0}L^{fcc}_{Ce:H,Va}$	−212.54	This work
CeH_x phase		
$G^{CeH_x}_{Ce:H:H}$	−57303 + 46.21*T + 0.25*GHSERCE + 0.375*GH2GAST	This work
$G^{CeH_x}_{Ce:Va:H}$	−843.16 - 0.028*T + 0.25*GHSERCE + 0.125*GH2GAST	This work
$G^{CeH_x}_{Ce:H:Va}$	−47144 + 34.18*T + 0.25*GHSERCE + 0.25*GH2GAST	This work
$G^{CeH_x}_{Ce:Va:Va}$	2706 + 5.97*T + 0.25*GHSERCE	This work
${}^{0}L^{CeH_x}_{Ce:Va:H,Va}$	−2868.8	This work
${}^{0}L^{CeH_x}_{Ce:H:H,Va}$	−1536.5	This work

Table 2
Assessed thermodynamic parameters for *model 2* of the Ce–H system [J/mol f.u.].

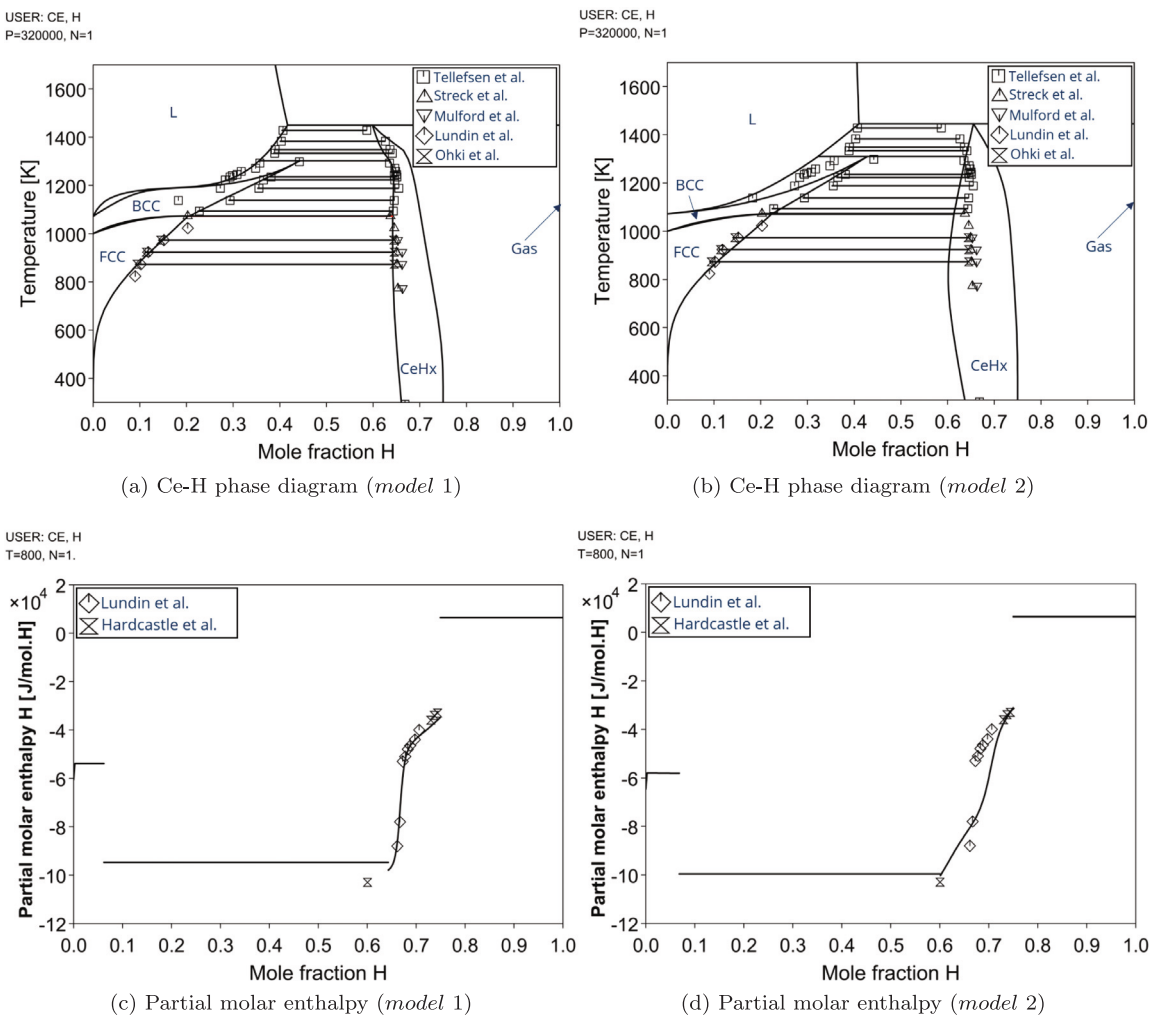
Parameter	Values	Reference
Liquid phase		
G^{liq}_{Ce}	GLIQCE	[37]
G^{liq}_{H}	+8035+25*T+2*T*ln(T) +0.5*GH2GAST	[41]
${}^{0}L^{liq}_{H,Ce}$	−33662 - 52.24*T	This work
${}^{1}L^{liq}_{H,Ce}$	−15188 + 40.31*T	This work
bcc phase		
$G^{bcc}_{Ce:Va}$	GBCCCE	[37]
$G^{bcc}_{Ce:H}$	−145710 + 127.45*T + GHSERCE + 1.5*GH2GAST	This work
fcc phase		
$G^{fcc}_{Ce:Va}$	GHSERCE	[37]
$G^{fcc}_{Ce:H}$	−59239 + 40.93*T + GHSERCE + 0.5*GH2GAST	This work
${}^{0}L^{fcc}_{Ce:H,Va}$	1.2782E+03	This work
CeH_x phase		
$G^{CeH_x}_{Ce:H:H}$	−60462 + 45.51*T + 0.25*GHSERCE + 0.375*GH2GAST	This work
$G^{CeH_x}_{Ce:Va:H}$	8392.9 + 0.25*GHSERCE + 0.125*GH2GAST	This work
$G^{CeH_x}_{Ce:H:Va}$	−50330 + 38.35*T + 0.25*GHSERCE + 0.25*GH2GAST	This work
$G^{CeH_x}_{Ce:Va:Va}$	8392.9 + 0.25*GHSERCE	This work
${}^{0}L^{CeH_x}_{Ce:Va:H,Va}$	−2373 - 0.62*T	This work
${}^{0}L^{CeH_x}_{Ce:H:H,Va}$	${}^{0}L^{CeH_x}_{Ce:Va:H,Va}$	This work
${}^{0}L^{CeH_x}_{Ce:H,Va:Va}$	−7518	This work

Table 3
DFT calculated ground state formation enthalpies (ΔE_f) of the unstable endmembers of *model 1* and the $CeNi_5H_7$ compound, alongside with the respective optimized CALPHAD energies (ΔH_f), normalized to one mole of atom.

Compound	Sublattice formula	ΔE_f [kJ/mol]	ΔH_f [kJ/mol]
Ce	(Ce)(Va)(Va)	8.454	10.082
CeH	$(Ce)(Va)_2(H)$	−1.983	−1.686
$CeNi_5H_7$	$(Ce)(Ni)_5(H)_7$	−11.176	−19.257

by the liquidus line of the bcc phase, the shape of the liquidus line becomes somewhat counterintuitive because it disagrees with other similar systems within the RE-H class [13,42]. The equilibrium calculation yields a sigmoidal shape of the liquidus line that approaches the solidus line around $x_H = 0.15$.

Although developed for describing ionic liquids, a two-sublattice model for the liquid phase was proposed by Zinkevich [42] to describe a similar metal-hydrogen system, which has shown more accurate results when describing metal-hydrogen systems that exhibit high solubility of hydrogen in the liquid phase. We thus propose that this model could be employed in future works when reassessing the Ce–H system



(a) Ce-H phase diagram (*model* 1)

(b) Ce-H phase diagram (*model* 2)

(c) Partial molar enthalpy (*model* 1)

(d) Partial molar enthalpy (*model* 2)

Fig. 2. Comparison of the calculated Ce–H phase diagrams and partial molar enthalpy at 800 K with experimental data from [16–20,26]. (*a*) and (*c*) using *model 1*; (*b*) and (*d*) using *model 2*.

as an essay to adjust the liquidus line to experimental points while maintaining a reasonable curvature.

In the phase diagram shown in Fig. 2(*a*) a pronounced reduction of the hydrogen equilibrium composition of the CeH_x phase above the peritectic temperature is observed. Although it agrees well with extracted compositions from plateau terminations of the PCI experiments, this experimental dataset should be expected to be relatively inaccurate due to its acquisition at higher temperatures and the presence of the liquid phase.

Furthermore, energy barriers inherent to phase transformations can also result in a shift in the observed experimental data value. The well-known Schwarz–Khachaturyan theory [50] and Flanagan theory [51] propose that the measured plateaux pressures during absorption-
desorption cycles are shifted from the bulk thermodynamic equilibrium due to induced elastic and inelastic stresses, respectively. While the Schwarz–Khachaturyan theory assumes coherent phase boundaries between the hydrogen-rich and hydrogen-poor phases, the Flanagan theory assumes incoherent phase boundaries and considers an additional energy barrier caused by the energy to form misfit dislocations. Lately, Weadock et al. proposed that these mechanisms should only result in an absolute shift of the equilibrium pressures and that a more plausible reason for the hysteresis effect should be primarily caused by interface pinning [22]. However, as the CALPHAD method presumes thermodynamic equilibrium of bulk phases, these effects must be disregarded. This is particularly important when integrating *ab initio* data within the assessments, as these data sources are only compatible if both represent bulk phase equilibria.

In *model 1*, the DFT calculated reaction enthalpy for the unstable end-members were chosen as starting values. During the optimization procedure, the DFT calculated $\Delta H^f_{\mathrm{Ce:Va:Va}}$, and $\Delta H^f_{\mathrm{Ce:Va:H}}$ values of 8.454 and −3.965 kJ/mol f.u. changed by 22 % and 18 %, respectively.

The partial molar enthalpies of hydrogen are plotted against the hydrogen content in Figs. 2(*c*) and 2(*d*) for *model 1* and *model 2*, respectively. Similar to the points representing phase equilibria, with more adjustable coefficients *model 1* demonstrated greater accuracy in describing partial molar enthalpies within the CeH_x single-phase region.

In the plateau region, the partial molar enthalpy of hydrogen is given the highest weight due to the relatively more abundant amount of experimental data available. Many authors reported similar values for the reaction enthalpy of 102 kJ/mol H [18–20], and 105 kJ/mol H [26]. Even though the adjustment of the partial molar enthalpy curve with the experimental dataset within the CeH_x phase region presents a higher RSD, the reaction enthalpy in the plateau region is calculated more accurately by the second approach (i.e., with fewer adjustable coefficients), demonstrating the compromise between the strategy for model complexity and accuracy.

In Fig. 3 the experimental PCI curves are plotted at 873 K and 1073 K superimposed to the pressure-composition phase diagrams calculated with the models from this work. It is worth mentioning that at 873 K the PCIs show a difference of one order of magnitude between Streck

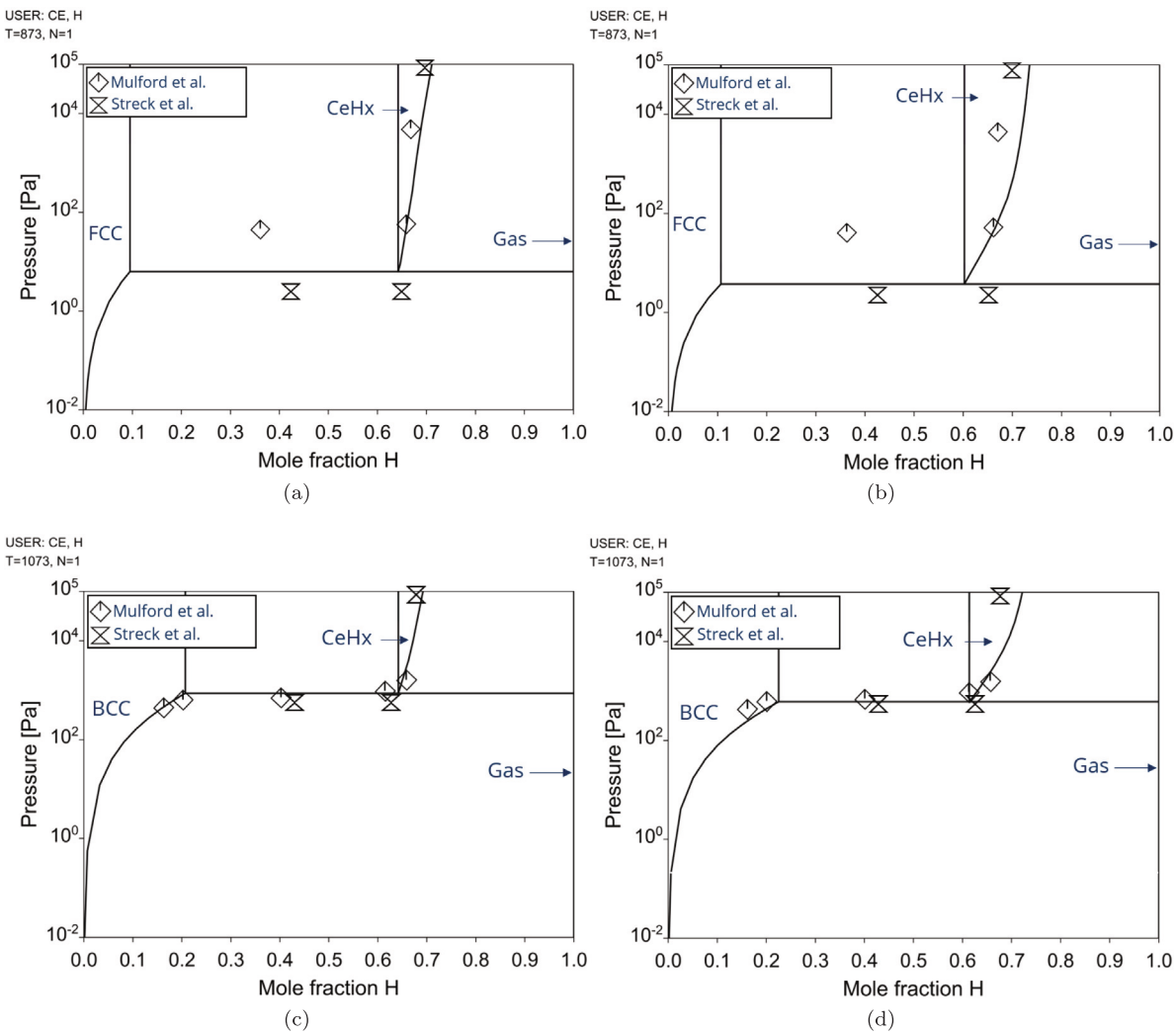


Fig. 3. Comparison of assessed PCI curve at 873 K and 1073 K with experimental data (a)+(c) with *model 1* (Table 1), (b)+(d) with *model 2* (Table 2). Note that the data at 873 K show a difference of one order of magnitude between the data from Mulford et al. [17] and Streck et al. [16].

et al. and Mulford et al. [16,17]. This emphasizes the difficulty of assuming bulk equilibrium data from PCI curves. Although these data were not included in the assessment, the models align closely with the experimental points. This good correspondence is a consequence of the great agreement obtained for the partial molar enthalpy of hydrogen, which acted as a modulator of the pressure dependence of the three-phase equilibria. These datasets were also more accurately represented by *model 1*, confirming its higher flexibility for adjusting to the experimental data, Whereas *model 2* yields slightly better agreement with the plateaux pressures. As the primary goal of a metal-hydride thermodynamic assessment is to identify the external conditions that lead to these three-phase equilibria, this comparison is of particular interest.

4.2. The $CeNi_5$–H system

In Fig. 4(*a*) the $CeNi_5$–H model is compared to the experimental PCI points. As available literature results are scarce, only a small amount of reported experimental data could be collected. The plateau terminations measured by the different authors vary about 5 at. %-H [8,35,36]. In contrast to the $LaNi_5$–H system, the compositional range of the plateau in the $CeNi_5$–H system is much smaller and varies from four to five hydrogen atoms per formula unit.

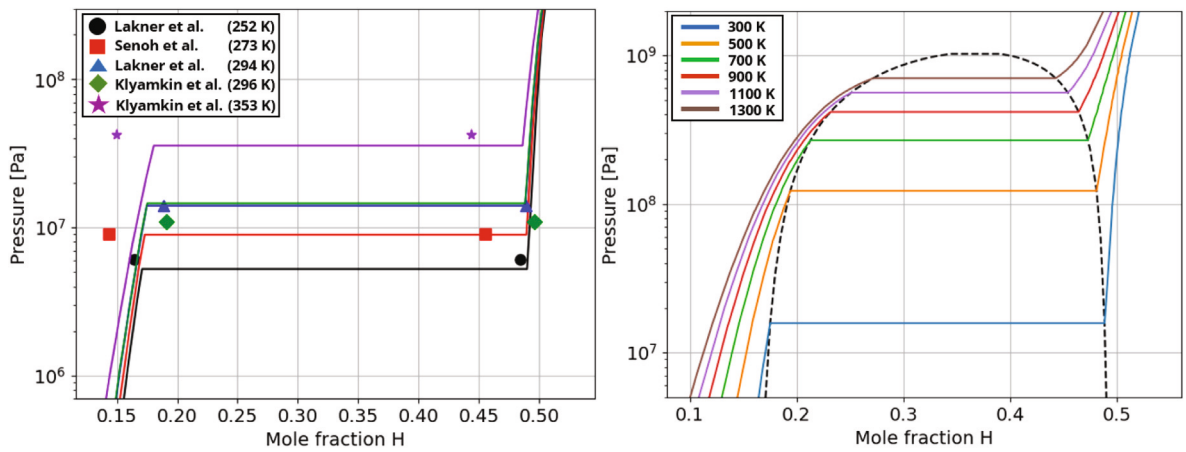
The enthalpy of formation of the hydride was calculated as $-250\,\mathrm{kJ\,mol^{-1}}$ using the thermodynamic model proposed, in comparison with the initial value of $-146\,\mathrm{kJ\,mol^{-1}}$ calculated through DFT. The hydride's formation entropy was found in the same order of magnitude as proposed in the literature [52] (see Table 4).

By applying the enthalpic and entropic terms for the end-members, only a single temperature-dependent parameter in the interaction term is needed to reproduce the experimental data. The inclusion of more interaction terms were tested, but did not improve the fitting results. While this is a drastic reduction of interaction parameters compared to the $LaNi_5$–H assessment of Palumbo et al. [13], the composition of the hydrogen-rich phase (β phase) in this sublattice model is stiff. This means the hydrogen content of the hydrogen-poor phase (α phase) in equilibrium with the β phase cannot be held constant, while the hydrogen content in the β phase is decreasing. The cause for this effect is that the miscibility gap between the α and β phases is solely expressed through the second-order interaction term ${}^{2}L^{CeNi_5H_x}_{Ce:Ni:H,Va}$. This provides a Gibbs energy of mixing ${}^{mix}G^{CeNi_5H_x}$ (i.e., ${}^{mix}G^{CeNi_5H_x} = {}^{id}G^{CeNi_5H_x} + {}^{ex}G^{CeNi_5H_x}$) that is symmetrical around $CeNi_5H_{3.5}$.

Experimental data for the $CeNi_5$–H system is in contrast to most of the other AB_5 systems [8,35,36]: The non-hydrogenated phase is calculated with constant solubility and composition slightly smaller than that of $CeNi_5H_1$ [35,53]. Herbst et al. calculated the internal equilibria for the $LaNi_5H_1$ and $LaNi_5H_7$ phase and showed a preference for hydrogen first occupying one 2b and then two 6c positions [54]. Even though not taken into account in this study, we propose that in future works, the $CeNi_5H_x$ phase could be modeled with two interstitial sublattices

Table 4
Assessed thermodynamic model parameters of the $CeNi_5H_x$ phase.

Parameter	Values	Reference
$CeNi_5H_x$ phase		
$G^{CeNi_5H_x}_{Ce:Ni:Va}$	−197460 + 24.31*T + GHSERCE + 5*GHSERNI	[27]
$G^{CeNi_5H_x}_{Ce:Ni:H}$	−250353 + 350.12*T + GHSERCE + 5*GHSERNI + 3.5*GH2GAST	This work
${}^2L^{CeNi_5H_x}_{Ce:Ni:H,Va}$	−214200	This work



(a) Comparison between the calculated PCI curves from this work (solid lines) and experimental data (○: 252 K [36], □: 273 K [35], △: 294 K [36], ◇: 296 K [8], ⋆: 353 K [8]).

(b) Calculated miscibility gap over temperature

Fig. 4. Calculated PCI curves with the assessed model for the $CeNi_5$–H system. Experimental data from the literature [8,35,36] are superimposed for comparison.

for the solubility of hydrogen and vacancies. The interstitial sublattices could correspond to the findings of Herbst et al. where the $LaNi_5H_1$ and $LaNi_5H_6$ compounds are on the convex hull, which leads to an 1:6 site ratio (relating to the 2b and $6c_1 6c_2$ sites), e.g. $(Ce)(Ni)_5(H, Va)_1(H, Va)_6$. Such a model would allow an equilibrium composition description without the need for a second-order hydrogen-vacancy interaction term to enforce the miscibility gap between α and β phases. However, it would come at the expense of acquiring formation enthalpies for the additional end-members that would be part of the model. Since these end-members formation energies can be obtained from first-principles methods, fewer excess terms would be required.

In Fig. 4(*b*) the results of the CALPHAD assessment calculated at high temperatures are shown. It illustrates an important aspect of using the ${}^2L^{CeNi_5H_x}_{Ce:Ni:H,Va}$ parameter: the ability to reduce the plateau compositional range with increasing temperature. This was experimentally validated in many AB_5 systems (A=La, Ce, Pr, Nd, Sm) for example by Senoh et al. [35]. Such a feature leads to a critical temperature from which the hydrogenation occurs continuously without a first-order phase transformation and agrees with other similar cases in the literature [49]. Note that the critical point presented in Fig. 4(*b*) is a metastable extrapolation because under experimental conditions the $CeNi_5$ phase would melt around 1600 K as shown in Fig. 1.

5. Conclusion

In this study assessments of the binary Ce–H and the ternary $CeNi_5$–H systems were performed. Literature data was collected and critically evaluated regarding accuracy and relevance for the thermodynamic description of the systems. Two different modeling approaches for the binary system were proposed leading to the following findings:

- Employing formation entropy terms for all end-members of the CeH_x hydride phase without the attribution of temperature-dependent terms within the hydrogen-vacancy interaction terms promoted a general better agreement with the experimental data while unusual curvatures in the phase boundary between bcc and liquid were observed. Here a two-sublattice model for the liquid phase has been proposed.
- Expressing temperature dependency with no entropy of formation terms for unstable end-members of the CeH_x hydride while attributing linear temperature dependence to the zero-th order hydrogen-vacancy interaction term promoted reasonable liquidus and solidus lines, aligning with other RE-H system's assessments.

The ternary $CeNi_5$–H system could be accurately described with fewer interaction parameters compared to CALPHAD models of similar AB_5 systems together with the evaluation of the plateaux terminations, pressures, and the critical point. For future models of the AB_5 system, a more comprehensive sublattice model based on DFT calculations has been proposed to adequately describe the plateau terminations of multicomponent AB_5 systems.

CRediT authorship contribution statement

Peter Hannappel: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ebert Alvares:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation. **Felix Heubner:** Writing – review & editing, Supervision, Conceptualization. **Claudio Pistidda:** Resources. **Paul Jerabek:** Writing – review & editing, Funding acquisition. **Thomas Weißgärber:** Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The Electronic Supplementary Information attached to this article contains the raw and processed data necessary to reproduce the findings.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.calphad.2024.102701.

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